National Testing Agency

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PGQP27

Group Number :	1
Group Id :	19088930
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Show Attended Group? :	No
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Break time :	0
Group Marks :	400
Is this Group for Examiner? :	No

PART A - General

Section Id :	19088979
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory

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Number of Questions :	25
Number of Questions to be attempted :	25
Section Marks :	100
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Sub-Section Number :	1
Sub-Section Id :	190889115
Question Shuffling Allowed :	Yes

Question Number : 1 Question Id : 1908893399 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Select the correct word that can best complete the given sentence:

By training children at a young age we can bring out the _____ talents in them.

(1) artificial (2) budding

(3) superficial (4) inherent

Options :

19088913501. 1

19088913502. 2

19088913503. 3

19088913504. 4

Question Number : 2 Question Id : 1908893400 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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From the choices given below, select the pair which exhibits the same relationship as the one in capitalized pair of words:

SOUP : APPETIZER

(1) coffee : bean

(2) pudding : dessert

(3) breakfast : cereal

(4) tea : drink

Options :

19088913505. 1

19088913506. 2

19088913507. 3

19088913508. 4

Question Number : 3 Question Id : 1908893401 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Select the most suitable synonym :

MUDDLE

(1) whisper

(2) horde

(3) disorder

(4) speculate

Options :

19088913509. 1

19088913510. 2

19088913511. 3

19088913512. 4

Question Number : 4 Question Id : 1908893402 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Identify the part of the body with which the disease is associated:

Leukemia

(1) heart
(2) skin
(3) lungs
(4) lymphatic system

Options :

19088913513. 1

19088913514. 2

19088913515. 3

19088913516. 4

Question Number : 5 Question Id : 1908893403 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Identify the meaning of the expression below from the options given:

Per se

(1) by word of mouth
(2) gossip
(3) by itself
(4) spontaneous

Options :

19088913517. 1

19088913518. 2

19088913519. 3

19088913520. 4

Question Number : 6 Question Id : 1908893404 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Identify the meaning of the underlined word:

Home appliances manufacturer 'B' electrical Limited is looking for <u>acquisitions</u> in the domestic market.

(1) obtainment
(2) apprentice
(3) attrition
(4) attention

Options :

19088913521. 1

19088913522. 2

19088913523. 3

19088913524. 4

Question Number : 7 Question Id : 1908893405 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Identify the meaning of the phrase below from the options given:

Leave to a person by a will

(1) beseech
(2) billow
(3) blanch
(4) bequeath

Options :

19088913525. 1

19088913526. 2

19088913527. 3

19088913528. 4

Question Number : 8 Question Id : 1908893406 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Select the most suitable antonym:

SKIMPY

(1) glaring
(2) modest
(3) affluent
(4) generous

Options :

19088913529. 1

19088913530. 2

19088913531. 3

19088913532. 4

Question Number : 9 Question Id : 1908893407 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Identify the meaning of the idiom from the options given:

A man of the world

(1) headstrong and arrogant
(2) highly trustworthy
(3) very popular because of success
(4) highly experienced in many fields

Options :

19088913533. 1

19088913534. 2

19088913535. 3

19088913536. 4

Question Number : 10 Question Id : 1908893408 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Select the correct word from the answer:

Now a days, it is difficult to _____ good books in English.

(1) track (2) verify

(3) find (4) know

Options :

19088913537. 1

19088913538. 2

19088913539. 3

19088913540. 4

Question Number : 11 Question Id : 1908893409 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

If Neena says "Anita's father Raman is the only son of my father-in-law Mahipal", then how is Bindu, who is the sister of Anita, related to Mahipal?

(1) Neice (2) Daughter

(3) Wife (4) None of these

Options :

19088913541. 1

19088913542. 2

19088913543. 3

19088913544. 4

Question Number : 12 Question Id : 1908893410 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Choose the missing term out of the given alternatives.

Y, W, U, S, Q, __, __

(1) N, J
(2) M, L
(3) J, R
(4) O, M

Options :

19088913545. 1

19088913546. 2

19088913547. 3

19088913548. 4

Question Number : 13 Question Id : 1908893411 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Arrange the given words in alphabetical order and tick the one that comes last.

(1) Abandon
(2) Actuate
(3) Accumulate
(4) Acquit

Options :

19088913549. 1

19088913550. 2

19088913551. 3

19088913552. 4

Question Number : 14 Question Id : 1908893412 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Raman ranks sixteenth from the top and forty nine from the bottom in a class. How many students are there in the class?

(1) 64
(2) 65
(3) 66
(4) None of these

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Options :

19088913553. 1

19088913554. 2

19088913555. 3

19088913556. 4

Question Number : 15 Question Id : 1908893413 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

If $\times$ means $\div$, $-$ means $\times$, $\div$ means $+$ and $+$ means $-$, then

$(3 - 15 \div 19) \times 8 + 6 = ?$

(1) 8 (2) 4

(3) 2 (4) -1

Options :

19088913557. 1

19088913558. 2

19088913559. 3

19088913560. 4

Question Number : 16 Question Id : 1908893414 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Choose the correct alternative based on relationship :

Botany : Plants::Entomology : ?

(1) Snakes (2) Insects

(3) Birds (4) Germs

Options :

19088913561. 1

19088913562. 2

19088913563. 3

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19088913564. 4

Question Number : 17 Question Id : 1908893415 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Choose the correct answer:

$9572 - 4018 - 2164 = ?$

(1) 3300 (2) 3390

(3) 3570 (4) 7718

Options :

19088913565. 1

19088913566. 2

19088913567. 3

19088913568. 4

Question Number : 18 Question Id : 1908893416 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The H.C.F. (Highest Common Factor) of 2923 and 3239 is :

(1) 37 (2) 47

(3) 73 (4) 79

Options :

19088913569. 1

19088913570. 2

19088913571. 3

19088913572. 4

Question Number : 19 Question Id : 1908893417 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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One fifth of a number exceeds one seventh of the same by 10. The number is :

(1) 125 (2) 150

(3) 175 (4) 200

Options :

19088913573. 1

19088913574. 2

19088913575. 3

19088913576. 4

Question Number : 20 Question Id : 1908893418 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

If a number, divided by 4, is reduced by 21, the number is :

(1) 18 (2) 20

(3) 28 (4) 38

Options :

19088913577. 1

19088913578. 2

19088913579. 3

19088913580. 4

Question Number : 21 Question Id : 1908893419 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The newly launched 100 rupees notes have the motif of ———— on the reverse side.

(1) Mangalyaan (2) Sanchi stupa

(3) Hampi with chariot (4) Rani ki vav

Options :

19088913581. 1

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19088913582. 2

19088913583. 3

19088913584. 4

Question Number : 22 Question Id : 1908893420 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The 'Mission Purvodaya' initiative is related to the development of which of the following?

(1) Agriculture sector
(2) Space sector
(3) Dairy sector
(4) Steel sector

Options :

19088913585. 1

19088913586. 2

19088913587. 3

19088913588. 4

Question Number : 23 Question Id : 1908893421 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Generally wooden doors are difficult to open or close during rainy season because of

(1) Diffusion
(2) Imbibition
(3) Osmosis
(4) Photosynthesis

Options :

19088913589. 1

19088913590. 2

19088913591. 3

19088913592. 4

Question Number : 24 Question Id : 1908893422 Question Type : MCQ Option Shuffling : No Is

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Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Mahabhasya, an outstanding work in the fields of Sanskrit grammar, is attributed to

(1) Ghosha
(2) Ashwins
(3) Patanjali
(4) Kalidas

Options :

19088913593. 1

19088913594. 2

19088913595. 3

19088913596. 4

Question Number : 25 Question Id : 1908893423 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Who among the following passed the Bengal Sati Regulation Act, 1829, which declared the practice of 'Sati' a punishable offence?

(1) Lord William Bentinck
(2) Lord Dalhousie
(3) Lord Wellesley
(4) Warren Hastings

Options :

19088913597. 1

19088913598. 2

19088913599. 3

19088913600. 4

PART B - MATHEMATICS

Section Id : 19088980

Section Number : 2

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Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	75
Number of Questions to be attempted :	75
Section Marks :	300
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Sub-Section Number :	1
Sub-Section Id :	190889116
Question Shuffling Allowed :	Yes

Question Number : 26 Question Id : 1908893424 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let G be a group of order 121, then

(1) G is non-abelian (2) G is cyclic

(3) Center of G has order 121 (4) None of these

Options :

19088913601. 1

19088913602. 2

19088913603. 3

19088913604. 4

Question Number : 27 Question Id : 1908893425 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The number of subgroups of $\mathbb{Z}_{48}$ is

(1) 10 (2) 48

(3) 2 (4) 100

Options :

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19088913605. 1

19088913606. 2

19088913607. 3

19088913608. 4

Question Number : 28 Question Id : 1908893426 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

How many normal subgroups does a non-abelian group G of order 21 have other than the identity subgroup $\{e\}$ and G?

(1) 0 (2) 7

(3) 3 (4) 1

Options :

19088913609. 1

19088913610. 2

19088913611. 3

19088913612. 4

Question Number : 29 Question Id : 1908893427 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let G be a finite abelian group of order n. Which one of the following is correct?

(1) If d divides n, there exists an element of order d in G

(2) If d divides n, there exists a subgroup of G of order d

(3) If every proper subgroup of G is cyclic then G is cyclic

(4) None of the above

Options :

19088913613. 1

19088913614. 2

19088913615. 3

19088913616. 4

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Question Number : 30 Question Id : 1908893428 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Total number of group homomorphisms from $\mathbb{Z}_5$ to $\mathbb{Z}_7$ is

(1) 35 (2) 7

(3) 5 (4) 1

Options :

19088913617. 1

19088913618. 2

19088913619. 3

19088913620. 4

Question Number : 31 Question Id : 1908893429 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The Ring M of 3×3 matrices with elements from the set of real numbers. The two operations in the ring M are usual addition and multiplication of matrices. Then M is a

(1) Commutative ring with zero divisors, without unity

(2) Non-commutative ring with zero divisors, with unity

(3) Commutative ring with unity

(4) Field

Options :

19088913621. 1

19088913622. 2

19088913623. 3

19088913624. 4

Question Number : 32 Question Id : 1908893430 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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The cardinality of a finite integral domain cannot be

(1) 21 (2) 7

(3) 5 (4) 3

Options :

19088913625. 1

19088913626. 2

19088913627. 3

19088913628. 4

Question Number : 33 Question Id : 1908893431 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is not prime ideal of the ring $\mathbb{Z}$ of integers ?

(1) $2\mathbb{Z}$ (2) $4\mathbb{Z}$

(3) $3\mathbb{Z}$ (4) $5\mathbb{Z}$

Options :

19088913629. 1

19088913630. 2

19088913631. 3

19088913632. 4

Question Number : 34 Question Id : 1908893432 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following sets of vectors $\alpha = (a_1, a_2, \ldots, a_n)$ in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n (n \geq 3)$?

(1) all α such that $a_1 \geq 0$ (2) all α such that $a_1 + 3a_2 = a_3$

(3) all α such that $a_2 = a_1^2$ (4) all α such that $a_1 a_2 = 0$

Options :

19088913633. 1

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19088913634. 2

19088913635. 3

19088913636. 4

Question Number : 35 Question Id : 1908893433 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The dimension of the vector space of all symmetric matrices of order $n \times n$ with real entries and trace equal to zero is

(1) $\frac{n(n+1)}{2}-1$

(2) $\frac{n(n-1)}{2}+1$

(3) $\frac{n(n+1)}{2}+1$

(4) $\frac{n(n-1)}{2}-1$

Options :

19088913637. 1

19088913638. 2

19088913639. 3

19088913640. 4

Question Number : 36 Question Id : 1908893434 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Suppose V is a finite dimensional non-zero vector space over the complex field $\mathbb{C}$ and $T : V \to V$ is a linear transformation such that range (T) = null space (T). Then which one of the following is correct ?

(1) The dimension of V may be 7

(2) The dimension of V may be 9

(3) The dimension of V may be 11

(4) The dimension of V may be 6

Options :

19088913641. 1

19088913642. 2

19088913643. 3

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19088913644. 4

Question Number : 37 Question Id : 1908893435 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $T:\mathbb{R}^2 \to \mathbb{R}^2$ be a linear operator defined by $T(x, y)=(x, 0)$. Then the matrix of T relative to the ordered basis $B=\{(0,1), (1,0)\}$ is

(1) $\begin{pmatrix}1 & 0\\ 1 & 0\end{pmatrix}$ (2) $\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}$

(3) $\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}$ (4) $\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$

Options :

19088913645. 1

19088913646. 2

19088913647. 3

19088913648. 4

Question Number : 38 Question Id : 1908893436 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let A and B be 2×2 matrices over the real field. Let $\det A$ and $\det B$ denote the determinants of the matrices A and B respectively. Then which one of the following is true ?

(1) $\det(A+B)+\det(A-B)=2\det A+2\det B$

(2) $\det(A+B)-\det(A-B)=2\det A-2\det B$

(3) $\det(A+B)+\det(A-B)=2\det A-2\det B$

(4) $\det(A+B)-\det(A-B)=2\det A+2\det B$

Options :

19088913649. 1

19088913650. 2

19088913651. 3

19088913652. 4

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Question Number : 39 Question Id : 1908893437 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $A=\begin{pmatrix}2 & 0 & 5\\ 1 & 2 & 3\\ -1 & 5 & 1\end{pmatrix}$. Let $X, Y \in \mathbb{R}^3$. The system of linear equations $AX = Y$ has a solution

(1) only for $Y=\begin{pmatrix}x\\0\\0\end{pmatrix}, x \in \mathbb{R}$

(2) for all $Y \in \mathbb{R}^3$

(3) only for $Y=\begin{pmatrix}0\\y\\z\end{pmatrix}, y, z \in \mathbb{R}$

(4) only for $Y=\begin{pmatrix}0\\y\\0\end{pmatrix}, y \in \mathbb{R}$

Options :

19088913653. 1

19088913654. 2

19088913655. 3

19088913656. 4

Question Number : 40 Question Id : 1908893438 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is correct ?

(1) $S = \{(1, 0, 0), (0, -1, 0), (1, 1, 0)\}$ is a linearly independent set of vectors in $\mathbb{R}^3$

(2) $S = \{(1, 0, 0), (0, 2, 0), (1, 1, 0)\}$ is a linearly independent set of vectors in $\mathbb{R}^3$

(3) A subset of a linearly dependent set of vectors is linearly independent

(4) A subset of a linearly independent set of vectors is linearly independent

Options :

19088913657. 1

19088913658. 2

19088913659. 3

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19088913660. 4

Question Number : 41 Question Id : 1908893439 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let W be a subspace of $\mathbb{R}^4$ given by $W = \{(x, y, z, w) : y + z + w = 0\}$. Then the dimension of W is

(1) 3 (2) 2

(3) 1 (4) 4

Options :

19088913661. 1

19088913662. 2

19088913663. 3

19088913664. 4

Question Number : 42 Question Id : 1908893440 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The eigenvalues of a skew-symmetric matrix are

(1) of absolute value 1 (2) real

(3) purely imaginary or zero (4) negative

Options :

19088913665. 1

19088913666. 2

19088913667. 3

19088913668. 4

Question Number : 43 Question Id : 1908893441 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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For the matrix $A=\begin{pmatrix}0 & 0 & 1\\ 2 & 1 & 1\\ 1 & 1 & 1\end{pmatrix}$, A^{-1} is given by

(1) $A^2+2A+3I$

(2) $A-3I$

(3) A^2-2A

(4) A^2-2A-I

Options :

19088913669. 1

19088913670. 2

19088913671. 3

19088913672. 4

Question Number : 44 Question Id : 1908893442 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let M^T denotes the transpose of M. Given that $M=\begin{pmatrix}\frac{3}{5} & \frac{3}{5}\\ x & \frac{3}{5}\end{pmatrix}$, where $M^T=M^{-1}$, then the value of x is

(1) $-\frac{4}{5}$

(2) 1

(3) 0

(4) $-\frac{3}{5}$

Options :

19088913673. 1

19088913674. 2

19088913675. 3

19088913676. 4

Question Number : 45 Question Id : 1908893443 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Let $\{a_n\}_{n=1}^{\infty}$ be a bounded sequence of real numbers. Then

(1) There is a subsequence of $\{a_n\}_{n=1}^{\infty}$ which is convergent

(2) Every subsequence of $\{a_n\}_{n=1}^{\infty}$ is convergent

(3) There is exactly one subsequence of $\{a_n\}_{n=1}^{\infty}$ which is convergent

(4) None of these

Options :

19088913677. 1

19088913678. 2

19088913679. 3

19088913680. 4

Question Number : 46 Question Id : 1908893444 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is incorrect ?

(1) Every subsequence of a convergent sequence of real numbers is convergent

(2) Every convergent sequence of real numbers is bounded

(3) Every bounded infinite set of real numbers has at least one limit point

(4) Every bounded sequence of real numbers is convergent

Options :

19088913681. 1

19088913682. 2

19088913683. 3

19088913684. 4

Question Number : 47 Question Id : 1908893445 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Which one of the following series is convergent ?

(1) $\sum_{n=1}^{\infty} n^2$ (2) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

(3) $\sum_{n=1}^{\infty} \left(\frac{3}{2}\right)^n$ (4) $\sum_{n=1}^{\infty} n^3$

Options :

19088913685. 1

19088913686. 2

19088913687. 3

19088913688. 4

Question Number : 48 Question Id : 1908893446 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $f:\mathbb{R} \to \mathbb{R}$ be a continuous function. If $\int_0^x f(2t)\,dt = \frac{x}{\pi}\sin(\pi x)$ for all $x \in \mathbb{R}$ then $f(2)$ is equal to

(1) 2 (2) 1

(3) 0 (4) –1

Options :

19088913689. 1

19088913690. 2

19088913691. 3

19088913692. 4

Question Number : 49 Question Id : 1908893447 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Which one of the following is incorrect ?

(1) A function which is uniformly continuous on an interval is continuous on that interval

(2) If a function is continuous in a closed and bounded interval, then it is bounded therein

(3) The function defined by $f(x)=\begin{cases} x\sin\frac{1}{x}, & when\ x\neq 0 \\ 0, & when\ x=0 \end{cases}$ is not continuous at $x=0$

(4) A function which is derivable at a point is necessarily continuous at that point

Options :

19088913693. 1

19088913694. 2

19088913695. 3

19088913696. 4

Question Number : 50 Question Id : 1908893448 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $\mathbb{Q}^c$ denotes the complement of the set of rational numbers $\mathbb{Q}$. The set of all boundary points of $\mathbb{Q}$ in the set of real numbers $\mathbb{R}$ is

(1) $\mathbb{Q}^c$ (2) $\mathbb{R}$

(3) ϕ (4) $\mathbb{Q}$

Options :

19088913697. 1

19088913698. 2

19088913699. 3

19088913700. 4

Question Number : 51 Question Id : 1908893449 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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The set $U = \{x \in \mathbb{R} \mid \sin x = \frac{1}{2}\}$ is

(1) closed

(2) open

(3) neither open nor closed

(4) both open and closed

Options :

19088913701. 1

19088913702. 2

19088913703. 3

19088913704. 4

Question Number : 52 Question Id : 1908893450 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

In the Taylor series expansion of e^x about $x = 2$, the coefficient of $(x-2)^4$ is

(1) $\frac{1}{256}$

(2) $\frac{2^4}{256}$

(3) $\frac{e^4}{256}$

(4) $\frac{e^2}{256}$

Options :

19088913705. 1

19088913706. 2

19088913707. 3

19088913708. 4

Question Number : 53 Question Id : 1908893451 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Let $x \in \mathbb{R}$. The set of all x at which the power series $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$ converges is

(1) $[-1, 1]$ (2) $(-1, 1)$

(3) $\mathbb{R}$ (4) $[0, 1]$

Options :

19088913709. 1

19088913710. 2

19088913711. 3

19088913712. 4

Question Number : 54 Question Id : 1908893452 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $(x, y) \in \mathbb{R}^2$. Which one of the following is true ?

(1) $\lim_{(x,y)\to(0,0)} \frac{2xy^2}{x^2+y^4}$ exists (2) $\lim_{(x,y)\to(0,0)} \frac{2xy^2}{x^2+y^4}$ does not exist

(3) $f(x\ y) = \frac{2xy^2}{x^2+y^4}$ is continuous at (0, 0) (4) None of these

Options :

19088913713. 1

19088913714. 2

19088913715. 3

19088913716. 4

Question Number : 55 Question Id : 1908893453 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Let $(x, y) \in \mathbb{R}^2$. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $f(x, y) = x^2 + 2xy + y^2$. Then

(1) The function f is a non-homogeneous function

(2) The function f is a homogeneous function of degree 1

(3) The function f is a homogeneous function of degree 2

(4) The function f is a homogeneous function of degree 0

Options :

19088913717. 1

19088913718. 2

19088913719. 3

19088913720. 4

Question Number : 56 Question Id : 1908893454 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

If $f: \mathbb{C} \to \mathbb{C}$ such that $f(z) = u + iv$ with $z = x + iy$. Then which one of the following is true for the function f to be analytic

(1) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x}$

(2) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

(3) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x}$ and $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial x}$

(4) $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 v}{\partial x \partial y}$

Options :

19088913721. 1

19088913722. 2

19088913723. 3

19088913724. 4

Question Number : 57 Question Id : 1908893455 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Which one of the following is an analytic function ?

(1) z (2) $\overline{z}$

(3) Real part of z (4) Imaginary part of z

Options :

19088913725. 1

19088913726. 2

19088913727. 3

19088913728. 4

Question Number : 58 Question Id : 1908893456 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The radius of convergence of the power series $\sum_{n=1}^{\infty} n^n z^n$ is

(1) 1 (2) ∞

(3) 0 (4) None of these

Options :

19088913729. 1

19088913730. 2

19088913731. 3

19088913732. 4

Question Number : 59 Question Id : 1908893457 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The maximum modulus of e^{z^2} on the set $S = \{z \in \mathbb{C} : 0 \le \mathrm{Re}(z) \le 1,\ 0 \le \mathrm{Im}(z) \le 1\}$ is

(1) e (2) 1

(3) $\frac{1}{e}$ (4) ∞

Options :

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19088913733. 1

19088913734. 2

19088913735. 3

19088913736. 4

Question Number : 60 Question Id : 1908893458 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is incorrect ?

(1) Every bounded entire function must be constant

(2) Every non-constant single-variable polynomial with complex coefficients has at least one complex root

(3) If $f(z)$ is analytic everywhere within a simply connected region D, then $\oint_C f(z)\,dz = 0$ for every simple closed path C lying in the region D

(4) None of these

Options :

19088913737. 1

19088913738. 2

19088913739. 3

19088913740. 4

Question Number : 61 Question Id : 1908893459 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The value of integral $\oint_{|z-i|=2} \frac{1}{z^2+4} dz$ is

(1) 0 (2) π

(3) $\frac{\pi}{2}$ (4) ∞

Options :

19088913741. 1

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19088913742. 2

19088913743. 3

19088913744. 4

Question Number : 62 Question Id : 1908893460 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $f : \mathbb{C} \to \mathbb{C}$ be an entire function. If real part of f is bounded then

(1) $f \equiv 0$

(2) f is nonzero constant

(3) Imaginary part of f is constant

(4) f is constant

Options :

19088913745. 1

19088913746. 2

19088913747. 3

19088913748. 4

Question Number : 63 Question Id : 1908893461 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is correct ?

(1) $S_1 = \{z \in \mathbb{C} : 1 < |z| < 2\}$ is connected

(2) $S_2 = \{z \in \mathbb{C} \ |z| < 1 \text{ and } |z-2| < 1\}$ is connected

(3) $S_3 = \{z \in \mathbb{C} : \text{Im}(z) > 1\}$ is not connected

(4) $S_4 = \{z \in \mathbb{C} : \text{Im}(z) = 1\}$ is not connected

Options :

19088913749. 1

19088913750. 2

19088913751. 3

19088913752. 4

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Question Number : 64 Question Id : 1908893462 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

About the definite integrals, which one of the following is incorrect ?

(1) $\int_a^b f(x)\,dx = -\int_a^b f(t)dt$

(2) $\int_a^b f(x)dt = -\int_b^a f(x)dx$

(3) If $a < c < b$, then $\int_a^b f(x)dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

(4) $\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & when\ f(x)\ is\ even \\ 0 \quad , & when\ f(x) is\ odd \end{cases}$

Options :

19088913753. 1

19088913754. 2

19088913755. 3

19088913756. 4

Question Number : 65 Question Id : 1908893463 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The value of integral $\int_{-\pi/2}^{\pi/2} x\cos x\,dx$ is

(1) $\pi - 2$

(2) $\pi + 2$

(3) 0

(4) $\frac{\pi}{2}$

Options :

19088913757. 1

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19088913758. 2

19088913759. 3

19088913760. 4

Question Number : 66 Question Id : 1908893464 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is incorrect?

(1) If a real valued function f is monotonic on $[a,b]$, then it is integrable on $[a,b]$

(2) A real valued bounded function f, having a finite number of points of discontinuity on $[a,b]$, is not integrable on $[a,b]$

(3) If a real valued function f is integrable on $[a,b]$, then f^2 is also integrable on $[a,b]$

(4) Every real valued continuous function f on $[a,b]$ is integrable on $[a,b]$

Options :

19088913761. 1

19088913762. 2

19088913763. 3

19088913764. 4

Question Number : 67 Question Id : 1908893465 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The value of integral $\int_0^{\pi} x \sin x \, dx$ is

(1) 2π (2) 0

(3) $\frac{\pi}{2}$ (4) π

Options :

19088913765. 1

19088913766. 2

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19088913767. 3

19088913768. 4

Question Number : 68 Question Id : 1908893466 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is the statement of fundamental theorem of calculus?

(1) Every continuous function is integrable

(2) Functions possessing primitives are necessarily continuous

(3) If a real valued function is bounded and integrable on $[a,b]$, and there exists a function F such that $\frac{d}{dx}F(x)=f$ on $[a,b]$, then $\int_a^b f(x)dx = F(b)-F(a)$

(4) Every monotonically increasing function is integrable

Options :

19088913769. 1

19088913770. 2

19088913771. 3

19088913772. 4

Question Number : 69 Question Id : 1908893467 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

After the change of order of integral, the double integral $\int_0^8 \int_{x^{\frac{1}{3}}}^2 dy\,dx$ becomes

(1) $\int_{x^{\frac{1}{3}}}^2 \int_0^8 dx\,dy$

(2) $\int_0^2 \int_0^{y^3} dx\,dy$

(3) $\int_8^0 \int_2^{x^{\frac{1}{3}}} dx\,dy$

(4) $\int_0^2 \int_{y^3}^0 dx\,dy$

Options :

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19088913773. 1

19088913774. 2

19088913775. 3

19088913776. 4

Question Number : 70 Question Id : 1908893468 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The area of the region bounded by the curves $x = y^2$ and $y = x^2$ is

(1) 1 (2) $\frac{2}{3}$

(3) $\frac{1}{3}$ (4) $\frac{4}{3}$

Options :

19088913777. 1

19088913778. 2

19088913779. 3

19088913780. 4

Question Number : 71 Question Id : 1908893469 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The value of the tripple integral $\int_{x=0}^{1}\int_{y=0}^{1}\int_{z=0}^{1}\left(x^2+y^2+z^2\right)dz\,dy\,dx$ is

(1) 0 (2) $-\frac{1}{3}$

(3) $-\frac{1}{4}$ (4) 1

Options :

19088913781. 1

19088913782. 2

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19088913783. 3

19088913784. 4

Question Number : 72 Question Id : 1908893470 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}=\frac{d^2y}{dx^2}$ is

(1) 1 (2) 2

(3) 3 (4) Not defined

Options :

19088913785. 1

19088913786. 2

19088913787. 3

19088913788. 4

Question Number : 73 Question Id : 1908893471 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The differential equation $x\frac{dx}{dy}+y=0$, represents a family of

(1) hyperbolas (2) exponential curves

(3) parabolas (4) circles

Options :

19088913789. 1

19088913790. 2

19088913791. 3

19088913792. 4

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Question Number : 74 Question Id : 1908893472 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The family of orthogonal trajectories of the family of parabolas $y = cx^2$ is

(1) $x^2 + 2y^2 = k^2$, where k is an arbitrary constant

(2) $2x^2 + y^2 = k^2$, where k is an arbitrary constant

(3) $x^2 + y^2 = k^2$, where k is an arbitrary constant

(4) $x^2 = ky$ where k is an arbitrary constant

Options :

19088913793. 1

19088913794. 2

19088913795. 3

19088913796. 4

Question Number : 75 Question Id : 1908893473 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

For $a,b,c \in \mathbb{R}$, if the differential equation $(ax^2 + bxy + y^2)dx + (2x^2 + cxy + y^2)dy = 0$ is exact, then

(1) $b = 2, c = 4$

(2) $a = b, c = 20$

(3) $b = 4, c = 2$

(4) $b = 2, a = 2c$

Options :

19088913797. 1

19088913798. 2

19088913799. 3

19088913800. 4

Question Number : 76 Question Id : 1908893474 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

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Correct Marks : 4 Wrong Marks : 1

Which one of the following is a linear differential equation?

(1) $\frac{dy}{dx} = \sin y$

(2) $y\frac{dy}{dx} = x$

(3) $\left(\frac{dy}{dx}\right)^2 = 1$

(4) $\frac{dy}{dx} = e^x$

Options :

19088913801. 1

19088913802. 2

19088913803. 3

19088913804. 4

Question Number : 77 Question Id : 1908893475 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is an integrating factor of the differential equation $ydx + 2xdy = 0$?

(1) $\mu(x,y) = x$

(2) $\mu(x,y) = y$

(3) $\mu(x,y) = 1$

(4) $\mu(x,y) = 2$

Options :

19088913805. 1

19088913806. 2

19088913807. 3

19088913808. 4

Question Number : 78 Question Id : 1908893476 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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The general solution of the differential equation $\frac{d^2y}{dx^2}+\frac{dy}{dx}-2y=0$ is

(1) $c_1e^{-2x}+c_2e^{x}$

(2) ce^{-2x}

(3) $c_1e^{x}+c_2x+c_3x^{3}$

(4) $c_1e^{2x}+c_2e^{x}$

Options :

19088913809. 1

19088913810. 2

19088913811. 3

19088913812. 4

Question Number : 79 Question Id : 1908893477 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The differential equation $\frac{d^2y}{dx^2}+\sin(x+y)=\sin x$ is

(1) Linear and non-homogeneous

(2) Linear and homogeneous

(3) Nonlinear and homogeneous

(4) Nonlinear and non-homogeneous

Options :

19088913813. 1

19088913814. 2

19088913815. 3

19088913816. 4

Question Number : 80 Question Id : 1908893478 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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The solution of the differential equation $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=0$ satisfying the conditions $y(0)=4$ and $\left.\frac{dy}{dx}\right|_{x=0}=8$ is

(1) e^{2x}

(2) $4e^{-2x}$

(3) $(4+16x)e^{-2x}$

(4) $4e^{2x}$

Options :

19088913817. 1

19088913818. 2

19088913819. 3

19088913820. 4

Question Number : 81 Question Id : 1908893479 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $y(x)$ be the solution of the initial value problem $x^2\frac{d^2y}{dx^2}-4x\frac{dy}{dx}+6y=0,\ x>0\ y(2)=0,$ $\left.\frac{dy}{dx}\right|_{x=2}=4$. Then value of $y(4)$ is

(1) 32

(2) 0

(3) 1

(4) $\frac{1}{32}$

Options :

19088913821. 1

19088913822. 2

19088913823. 3

19088913824. 4

Question Number : 82 Question Id : 1908893480 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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$y = Ae^{2x} + Be^{-2x}$, where A and B are arbitrary constants, is a solution of

(1) $B\frac{d^2y}{dx^2} + Ay = 0$

(2) $A\frac{d^2y}{dx^2} + 4y = 0$

(3) $\frac{d^2y}{dx^2} + 2y = 4$

(4) $\frac{d^2y}{dx^2} - 4y = 0$

Options :

19088913825. 1

19088913826. 2

19088913827. 3

19088913828. 4

Question Number : 83 Question Id : 1908893481 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let P and Q be real constants. Then $y = e^{mx}$ is a solution of differential equation $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$, if

(1) $m^2 + Pm - Q = 0$

(2) $m^2 + Pm + Q = 0$

(3) $m^2 - Pm + Q = 0$

(4) $m^2 - Pm - Q = 0$

Options :

19088913829. 1

19088913830. 2

19088913831. 3

19088913832. 4

Question Number : 84 Question Id : 1908893482 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Let $y_1(x)$ and $y_2(x)$, defined on $[0,1]$, be twice continuously differentiable functions satisfying $\frac{d^2y}{dx^2}+\frac{dy}{dx}+y=0$. Let $W(x)$ be Wronskian of y_1 and y_2 which satisfies $W\left(\frac{1}{2}\right)=0$.

Then

(1) $W(x)\neq 0$ for all $x\in\left[0,\frac{1}{2}\right)\cup\left(\frac{1}{2},1\right]$

(2) $W(x)>0$ for all $x\in\left[\frac{1}{2},1\right]$

(3) $W(x)=0$ for all $x\in[0,1]$

(4) $W(x)<0$ for all $x\in\left[0,\frac{1}{2}\right)$

Options :

19088913833. 1

19088913834. 2

19088913835. 3

19088913836. 4

Question Number : 85 Question Id : 1908893483 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The two linearly independent solutions of the differential equation $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=0$ are

(1) e^x and $2e^x$

(2) x and $5x^2$

(3) $\sin x$ and $9\sin x$

(4) e^x and xe^x

Options :

19088913837. 1

19088913838. 2

19088913839. 3

19088913840. 4

Question Number : 86 Question Id : 1908893484 Question Type : MCQ Option Shuffling : No Is

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Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\nabla = \frac{\partial}{\partial x} + \hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$. If $\phi(x, y, z)$ is a solution of the Laplace's equation then the vector field $\left(\nabla\phi + \vec{r}\right)$ is

(1) solenoidal but not irrotational

(2) neither solenoidal nor irrotational

(3) irrotational but not solenoidal

(4) both solenoidal and irrotational

Options :

19088913841. 1

19088913842. 2

19088913843. 3

19088913844. 4

Question Number : 87 Question Id : 1908893485 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let ϕ and $\vec{f}$ be differentiable scalar and vector functions, respectively and both have continuous second partial derivatives. Then which one of the following is true?

(1) The curl of the gradient of ϕ is never zero

(2) The divergence of the curl of $\vec{f}$ is zero

(3) $\vec{f}$ is irrotational if its divergence is zero

(4) $\vec{f}$ is called solenoidal if curl of $\vec{f}$ is zero

Options :

19088913845. 1

19088913846. 2

19088913847. 3

19088913848. 4

Question Number : 88 Question Id : 1908893486 Question Type : MCQ Option Shuffling : No Is

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Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The magnitude of the gradient of function $f : \mathbb{R}^3 \to \mathbb{R}$ such that $f(x,y,z) = xyz^2$ at the point $(1, 0, 2)$ is

(1) 4 (2) 3

(3) 1 (4) 0

Options :

19088913849. 1

19088913850. 2

19088913851. 3

19088913852. 4

Question Number : 89 Question Id : 1908893487 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Suppose $\vec{r} = x\hat{i} + y\hat{j}$ be a position vector of the point $P(x,y)$. Let $\vec{f} = -3x^2\hat{i} + 5xy\hat{j}$ and let C be the curve $y = 2x^2$ in the xy-plane. Then the value of the line integral $\int_C \vec{f} \cdot d\vec{r}$ from the point $P_1(0, 0)$ to $P_2(1, 2)$ is

(1) 0 (2) $\frac{1}{7}$

(3) 1 (4) 7

Options :

19088913853. 1

19088913854. 2

19088913855. 3

19088913856. 4

Question Number : 90 Question Id : 1908893488 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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If $\vec{F}$ is a conservative vector field, then

(1) $\vec{F}$ is solenoidal

(2) curl of $\vec{F}$ is nonzero

(3) $\vec{F}$ is irrotational

(4) divergence of $\vec{F}$ is zero

Options :

19088913857. 1

19088913858. 2

19088913859. 3

19088913860. 4

Question Number : 91 Question Id : 1908893489 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Let $\vec{F} = 2xz\hat{i} - x\hat{j} + y^2\hat{k}$. The value of $\iiint_V \vec{F}\,dV$, where V is the region bounded by the surfaces $x = 0, y = 0, y = 6, z = x^2, z = 4$ is

(1) 1

(2) $8\hat{i} + 4\hat{j} + 4\hat{k}$

(3) $128\hat{i} - 24\hat{j} + 385\hat{k}$

(4) $128\hat{i} - 24\hat{j} + 384\hat{k}$

Options :

19088913861. 1

19088913862. 2

19088913863. 3

19088913864. 4

Question Number : 92 Question Id : 1908893490 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Which one of the following is true?

(1) Green's theorem in the plane is a special case of Stokes' theorem

(2) Stokes' theorem is a special case of Green's theorem

(3) Fundamental theorem of integral calculus is the generalization of Green's theorem in plane

(4) Green's theorem in plane is the generalization of Gauss' divergence theorem

Options :

19088913865. 1

19088913866. 2

19088913867. 3

19088913868. 4

Question Number : 93 Question Id : 1908893491 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is incorrect?

(1) The intersection of two convex sets is a convex set

(2) The intersection of any finite number of convex sets is a convex set

(3) The set $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ is a convex set

(4) The union of two convex sets is also a convex set

Options :

19088913869. 1

19088913870. 2

19088913871. 3

19088913872. 4

Question Number : 94 Question Id : 1908893492 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Let $S=\{(x, y)\in \mathbb{R}^2: x^2+y^2=1\}$ and $T=\{(x,y)\in\mathbb{R}^2 : y\geq x^2\}$. Then which one of the following is correct

(1) S is a convex set but T is not a convex set

(2) T is a convex set but S is not a convex set

(3) Both S and T are not convex sets

(4) Both S and T are convext sets

Options :

19088913873. 1

19088913874. 2

19088913875. 3

19088913876. 4

Question Number : 95 Question Id : 1908893493 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is in the convex hull of the points (0, 1), (1, 0) and (1, 1)?

(1) (0, 0)

(2) $\left(0, \frac{3}{2}\right)$

(3) $\left(\frac{1}{2}, \frac{1}{2}\right)$

(4) (–10, 10)

Options :

19088913877. 1

19088913878. 2

19088913879. 3

19088913880. 4

Question Number : 96 Question Id : 1908893494 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

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Let $S=\{(x,y)\in\mathbb{R}^2: x^2+y^2<1\}$ and $T=\left\{(0,1),\left(\frac{1}{2},\frac{1}{2}\right),(1,0)\right\}$, then the convex hull of $S\cup T$ is

(1) $\{(x,y)\in\mathbb{R}^2: x^2+y^2\leq 1\}$

(2) $S\cup T$

(3) $\{(x,y)\in\mathbb{R}^2: x^2+y^2<1\}$

(4) $S\cap T$

Options :

19088913881. 1

19088913882. 2

19088913883. 3

19088913884. 4

Question Number : 97 Question Id : 1908893495 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

Which one of the following is true?

(1) A boundary point of a convex set is the extreme point of the convex set

(2) An extreme point of a convex set is the boundary point of the convex set

(3) Convex hull of a set $S\subset\mathbb{R}^n$ is the largest convex set containing S

(4) A hyperplane in $\mathbb{R}^n$ is not a convex set

Options :

19088913885. 1

19088913886. 2

19088913887. 3

19088913888. 4

Question Number : 98 Question Id : 1908893496 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

One of the vertex of the convex set $\{(x,y)\in\mathbb{R}^2: x+2y\geq 2, 2x+3y\leq 6, x\geq 0, y\geq 0\}$ is

(1) (0, 0)

(2) (1, 1)

(3) (2, 0.5)

(4) (3, 0)

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Options :

19088913889. 1

19088913890. 2

19088913891. 3

19088913892. 4

Question Number : 99 Question Id : 1908893497 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The value of the objective function at an optimal solution of the linear programming problem $\min z = x_1 + x_2$ subject to the conditions $x_1 - x_2 = -5,\ x_1 \geq 0,\ x_2 \geq 0$ will be

(1) 0 (2) 10

(3) 5 (4) –5

Options :

19088913893. 1

19088913894. 2

19088913895. 3

19088913896. 4

Question Number : 100 Question Id : 1908893498 Question Type : MCQ Option Shuffling : No Is Question Mandatory : No

Correct Marks : 4 Wrong Marks : 1

The linear programming problem $\max z = -x_1 + 2x_2$ subject to conditions $-x_1 + x_2 \leq 1,$ $-x_1 + 2x_2 \leq 4,\ 0 \leq x_1 \leq 5, x_2 \geq 0$ has

(1) multiple optimal solutions

(2) unique optimal solution

(3) no solution

(4) unbounded solution

Options :

19088913897. 1

19088913898. 2

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19088913899. 3

19088913900. 4

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Final Answer Keys on which score compiled on 20.10.2021

Exam Date : 15.09.2021 Shift : Shift 2 Test Paper Code : PGQP27

Question ID	Correct Option ID	Question ID	Correct Option ID	Question ID	Correct Option ID
1908893399	19088913504	1908893442	19088913673	1908893484	19088913843
1908893400	19088913506	1908893443	19088913677	1908893485	19088913846
1908893401	19088913511	1908893444	19088913684	1908893486	19088913849
1908893402	19088913516	1908893445	19088913686	1908893487	19088913856
1908893403	19088913519	1908893446	19088913692	1908893488	19088913859
1908893404	19088913521	1908893447	19088913695	1908893489	19088913864
1908893405	19088913528	1908893448	19088913698	1908893490	19088913865
1908893406	19088913530	1908893449	19088913701	1908893491	19088913872
1908893407	19088913536	1908893450	19088913708	1908893492	19088913874
1908893408	19088913539	1908893451	19088913711	1908893493	19088913879
1908893409	19088913544	1908893452	19088913714	1908893494	19088913882
1908893410	19088913548	1908893453	19088913719	1908893495	19088913886
1908893411	19088913550	1908893454	19088913722	1908893496	19088913892
1908893412	19088913553	1908893455	19088913725	1908893497	19088913895
1908893413	19088913559	1908893456	19088913731	1908893498	19088913897
1908893414	19088913562	1908893457	19088913733		
1908893415	19088913566	1908893458	19088913740		
1908893416	19088913572	1908893459	19088913743		
1908893417	19088913575	1908893460	19088913748		
1908893418	19088913579	1908893461	19088913749		
1908893419	19088913584	1908893462	19088913753, 19088913754		
1908893420	19088913588	1908893463	19088913759		
1908893421	19088913590	1908893464	19088913762		
1908893422	19088913595	1908893465	19088913768		
1908893423	19088913597	1908893466	19088913771		
1908893424	19088913603	1908893467	19088913774		
1908893425	19088913605	1908893468	19088913779		
1908893426	19088913612	1908893469	19088913784		
1908893427	19088913614	1908893470	19088913786		
1908893428	19088913620	1908893471	19088913792		
1908893429	19088913622	1908893472	19088913793		
1908893430	19088913625	1908893473	19088913799		
1908893431	19088913630	1908893474	19088913804		
1908893432	19088913634	1908893475	19088913806		
1908893433	19088913637	1908893476	19088913809		
1908893434	19088913644	1908893477	19088913816		
1908893435	19088913646	1908893478	19088913819		
1908893436	19088913649	1908893479	19088913821		
1908893437	19088913654	1908893480	19088913828		
1908893438	19088913660	1908893481	19088913830		
1908893439	19088913661	1908893482	19088913835		
1908893440	19088913667	1908893483	19088913840		
1908893441	19088913672				

Note :- Correct Option ID 'Dropped' means the question is Dropped

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